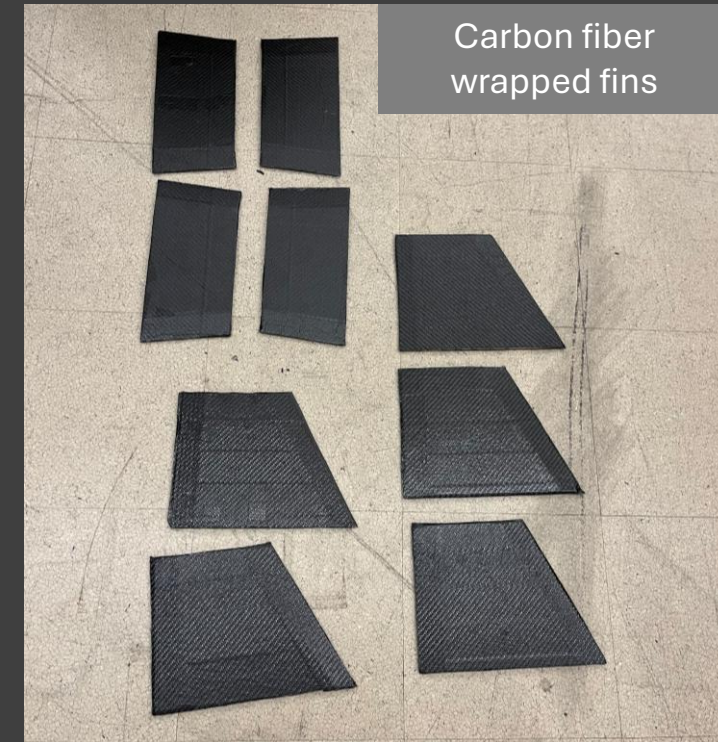
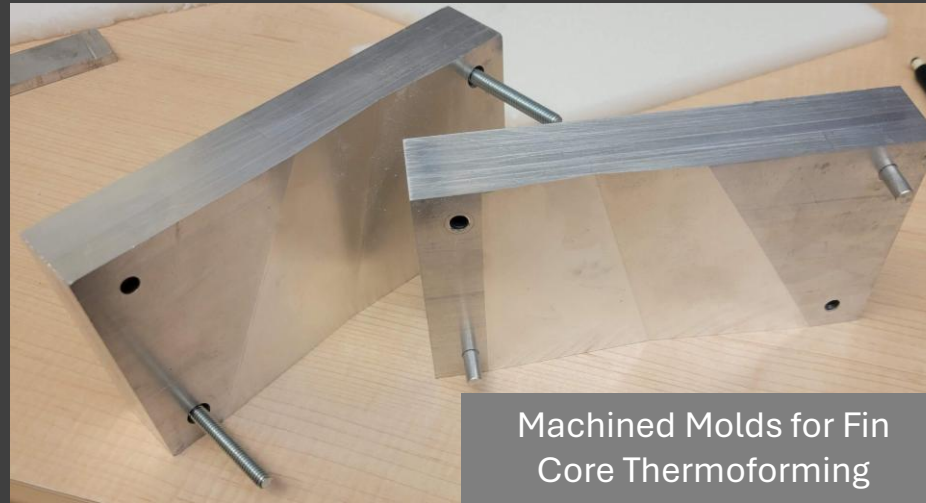


Engineering Portfolio

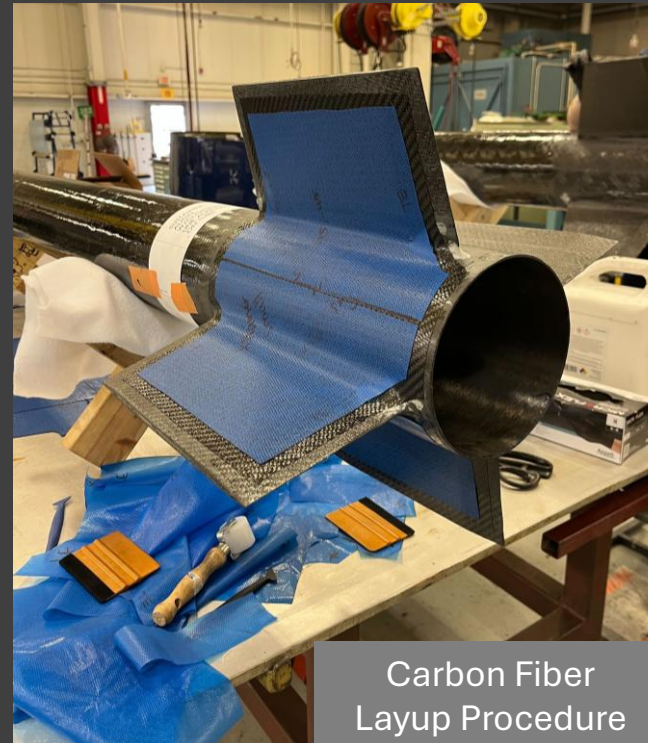
Nathaniel Murcin

GTXR Fin Team Lead

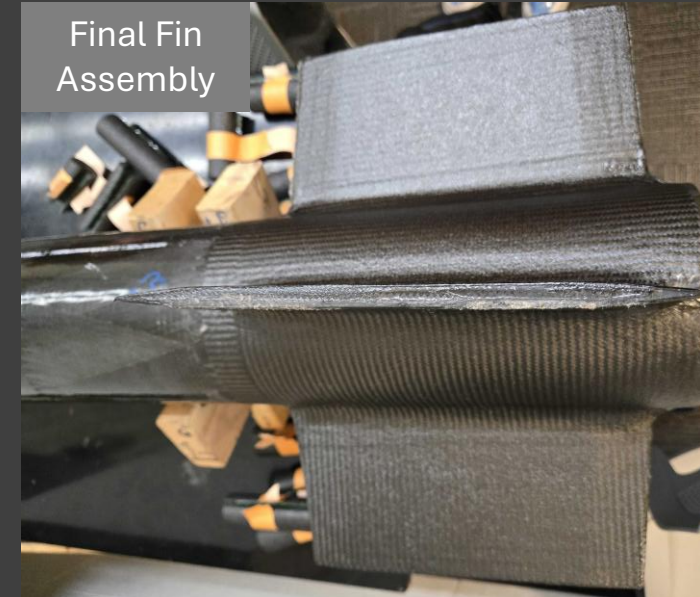
- 2 years of design/iteration of 2 stage hypersonic sounding rocket
- My contributions
 - Performed high-level design
 - Tested Equipment & Materials Research
 - Flight Hardware
 - Evaluated manufacturing methods
 - Create model to calculate flutter speed
 - Calculated Cant Angle



Vacuum bagging
for oven curing

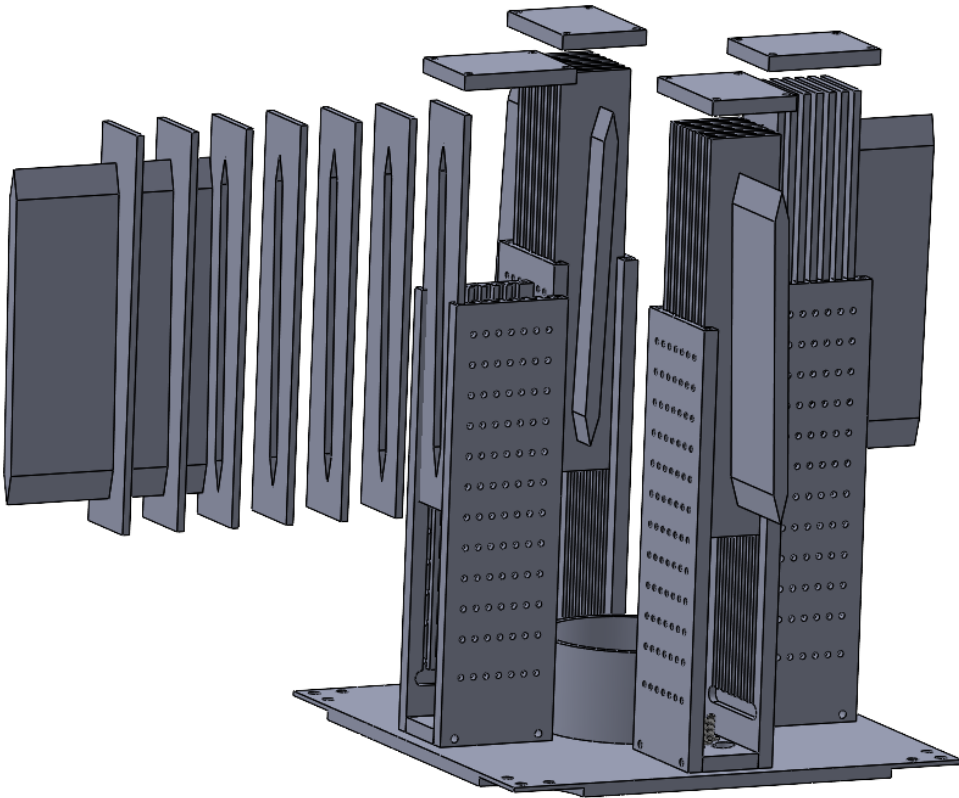


Final Fin
Assembly



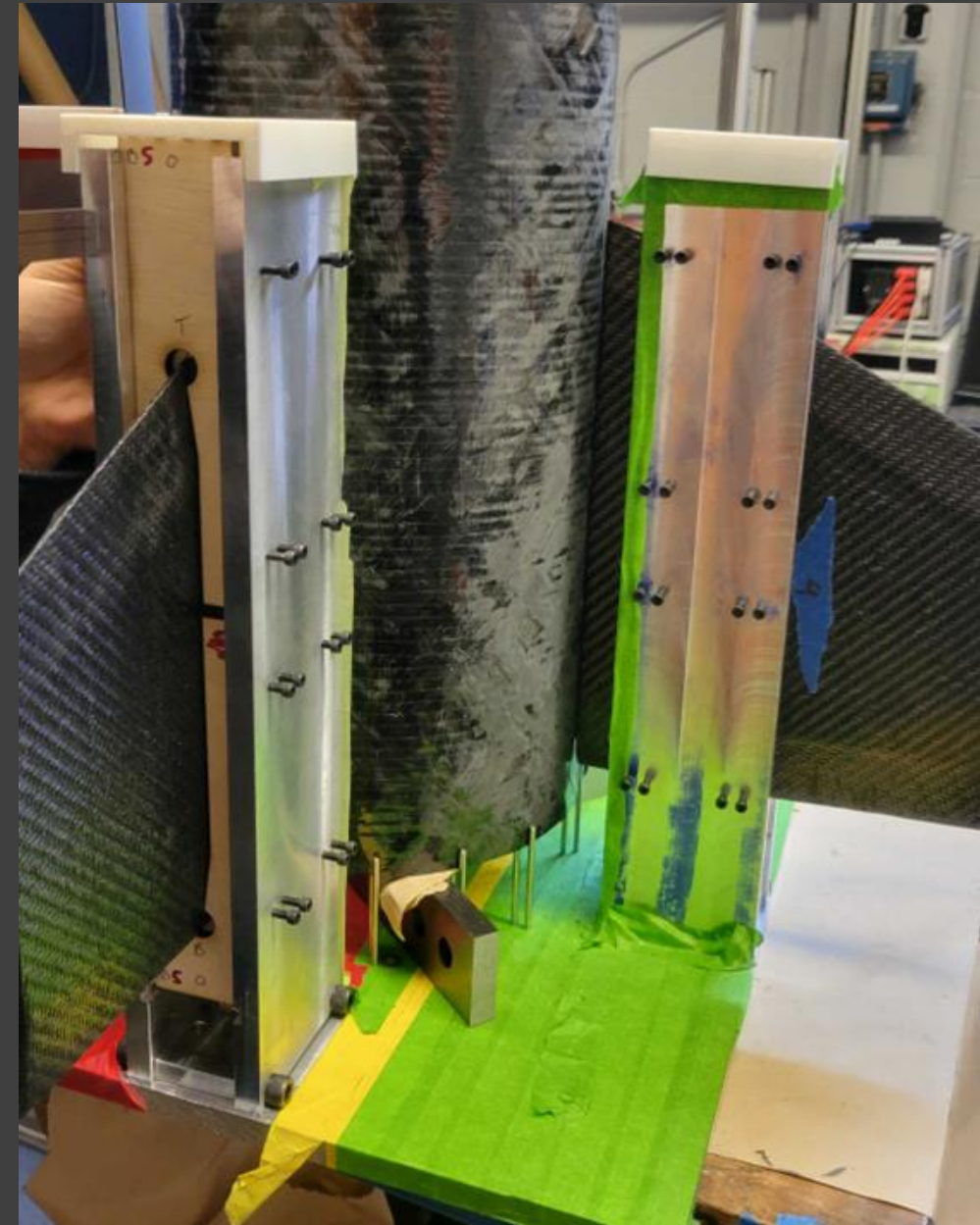
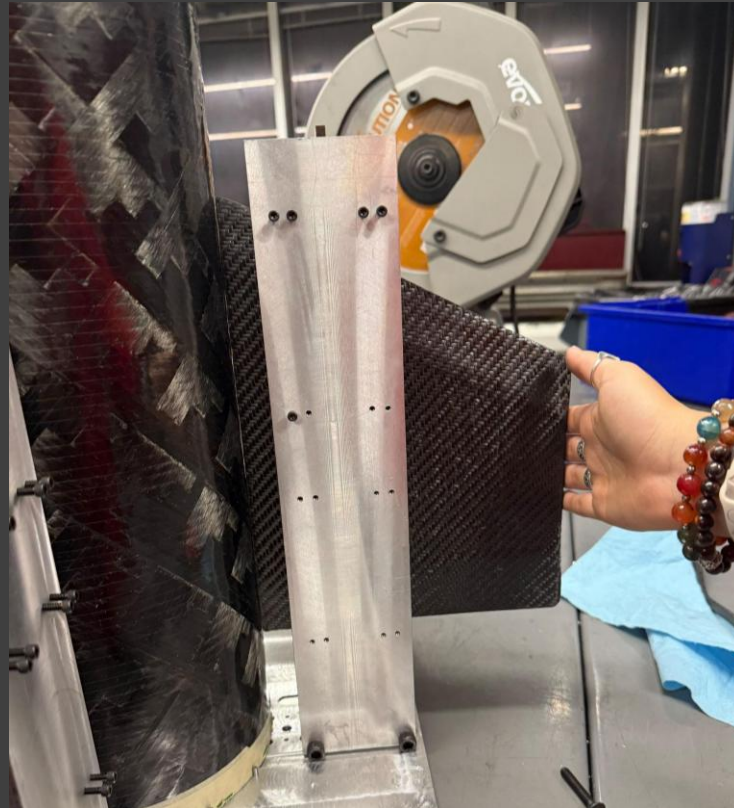
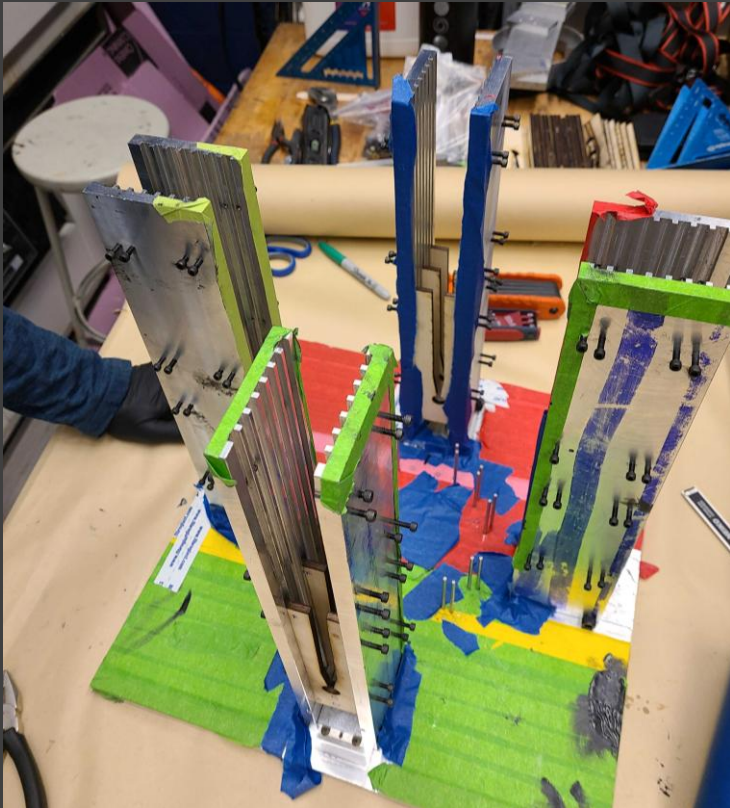
Fin Alignment Jig

- Identified fin cant angle required to address roll coupling failure.
- Generated CAD design for alignment jig
- Machined aluminum components



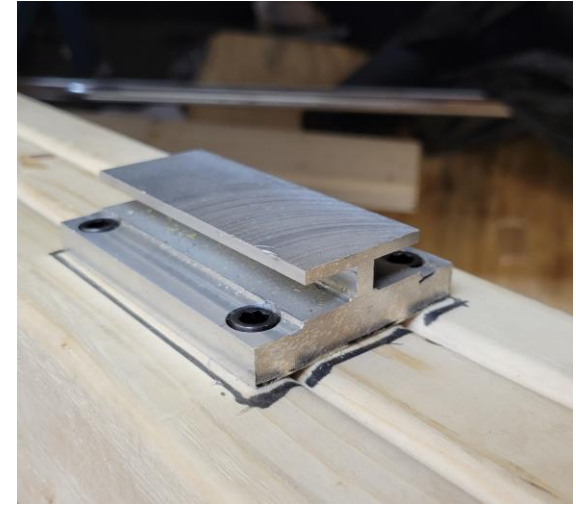
Jig Utilization for Fin Alignment

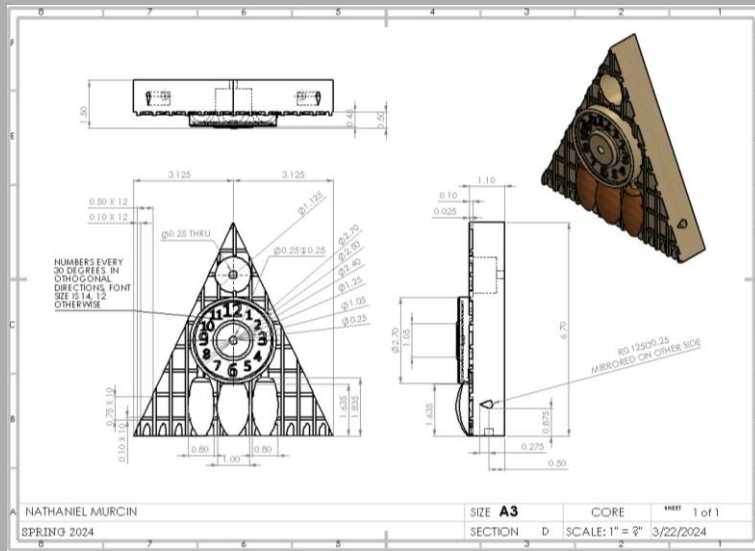
- $< 0.1^\circ$ tolerances
- Fully adjustable to all future designs
- Achieves desired fin cant angle to spin stabilize



Mock Launch Rail

- The 2024-2025 GTXR rocket architecture required a modified launch rail.
- Within 2 weeks I completed the following:
 - bending/buckling calculations
 - CAD modeling of components
 - manufactured all wood, steel, and aluminum components
- Resulted in functional assembled rail for mock test

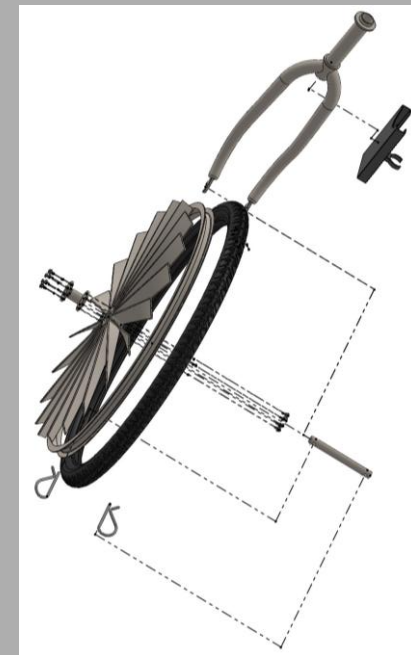




Cuckoo Clock – Personal Project

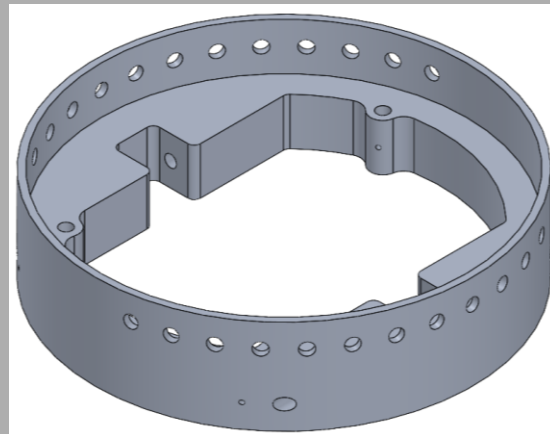


Turbine Electric Bike – Team Project

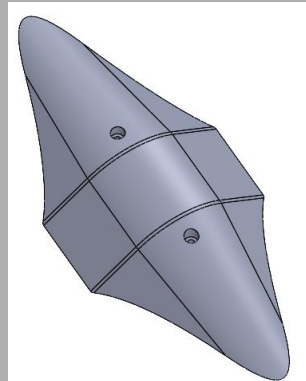


Rocket Components

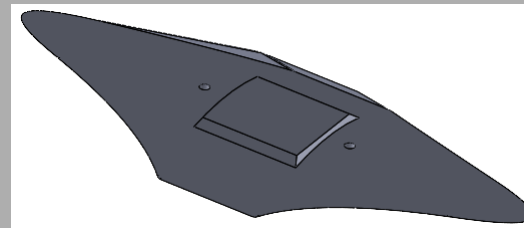
SolidWorks Samples



Parachute Ring



Sustainer
Shroud



Alignment Jig
Tower Base

